AgriChat: A Hybrid Structured–Unstructured RAG System for Automated Analysis of Agricultural Variety Trial Reports

Kaushik Sivakumar
kaushiks5499@tamu.edu

Prachi Dudhe
pra24@tamu.edu

Venkateswarlu Nagineni
kaushiks5499@tamu.edu

***Abstract*—Agricultural trial reports contain critical performance measurements for crop varieties, yet these documents are predominantly distributed as unstructured PDFs, making large-scale comparison and analysis difficult. This work addresses the need for an automated system capable of extracting, structuring, and interpreting agronomic information directly from heterogeneous field trial documents. The objective of this study is to develop a hybrid retrieval-augmented generation (RAG) framework that supports both semantic querying and structured analysis of cotton variety performance data.**

We present AgriChat, an end-to-end pipeline that converts PDF trial reports into structured rows through page-level text extraction, table normalization, header inference, numeric field validation, and metadata reconstruction. Unstructured text blocks and structured trial rows are embedded using a domain-adapted encoder and stored in a persistent vector index via LlamaIndex. A dual-mode query router integrates semantic retrieval with rule-based detection of varieties, counties, and years, enabling both free-form RAG responses and structured comparative outputs. The system incorporates a high-throughput LLM backend optimized via Groq acceleration to produce analytical summaries.

Experimental evaluations show that AgriChat achieves high retrieval consistency and successfully resolves comparison queries across multiple counties and years. The results demonstrate that combining structured normalization with unstructured RAG yields significantly improved interpretability and domain reliability. The proposed framework establishes a scalable foundation for automated decision support systems in agricultural analytics.

I. Introduction

Agricultural field trials play a central role in evaluating crop varieties, quantifying fiber quality, and guiding region-specific planting recommendations. These trials are commonly published as PDF reports containing a mixture of narrative text, tabular summaries, and region-specific experimental metadata. Despite their importance, the majority of these documents remain unstructured and manually curated, limiting their integration into digital decision-support systems. Recent advances in large language models (LLMs) and retrieval-augmented generation (RAG) frameworks have enabled automated knowledge extraction from heterogeneous corpora [1], [2], yet their application to agronomic data remains underexplored.

Existing approaches for agricultural data digitization primarily rely on optical character recognition (OCR), fixed-schema table parsers, or supervised extraction models tailored to specific document layouts [3], [4]. However, field trial documents exhibit substantial structural variability, including inconsistent table headers, mixed measurement units, ambiguous abbreviations, and nested metadata. These inconsistencies reduce the reliability of conventional text-mining pipelines. Furthermore, most RAG systems designed for general domains do not incorporate structured numeric data or region-specific agricultural terminology, resulting in retrieval errors, incomplete comparisons, or hallucinated interpretations when applied to crop performance analysis [5].

These limitations highlight a critical gap: the absence of an automated, domain-adaptive system capable of converting agricultural trial PDFs into structured, queryable datasets and supporting analytical reasoning over both text and numeric performance metrics. Addressing this gap is essential for enabling scalable, evidence-based decision making among breeders, agronomists, and extension specialists.

The motivation for this work arises from the need to (i) reduce the manual effort required to interpret variety trial documents, (ii) provide consistent cross-county and cross-year comparisons, and (iii) enable context-aware retrieval that integrates domain-specific numeric metrics such as yield, turnout, micronaire, and fiber strength. Traditional dense retrieval approaches cannot meet these requirements due to their reliance on unstructured text alone, while purely structured pipelines lack semantic flexibility.

This study aims to design and evaluate a hybrid system that unifies structured data extraction with unstructured semantic retrieval using a domain-adapted RAG architecture. The proposed system, **AgriChat**, processes PDF trial reports through a multi-stage pipeline involving page-level extraction, schema normalization, embedding generation, vector indexing, and LLM-based reasoning. The system further incorporates a rule-driven query router to support comparative variety analysis by detecting varieties, counties, and years directly from user queries.

The main contributions of this work are as follows:

- **A robust extraction and normalization pipeline** for converting heterogeneous PDF field trial reports into standardized row-level datasets.
- **A persistence-backed vector indexing module** integrating structured and unstructured embeddings using LlamaIndex.
- **A hybrid RAG-based query engine** capable of switching between structured dataframe analysis and unstruc-

tured semantic retrieval using variety–year–county routing logic.

- **An accelerated LLM reasoning layer** powered by Groq hardware to generate analytical summaries with low latency.
- **A complete interactive system, AgriChat**, that enables users to perform semantic search, structured comparison, and domain-specific reasoning over cotton variety trial data.

II. Literature Review

Agricultural variety trial reports are primarily distributed as unstructured PDFs, creating substantial challenges for automated extraction and analysis. Traditional digitization workflows rely on OCR pipelines, fixed-schema table parsers, or layout-specific extraction models, yet prior studies show that table structures in scientific and agronomic documents vary widely in formatting, header placement, and metadata representation [3], [4]. These inconsistencies limit the reliability of single-stage extraction approaches.

Recent work in document intelligence highlights advances in table detection, header inference, and structure reconstruction using deep learning methods [6], [7]. However, such systems remain sensitive to irregular table structures, ambiguous abbreviations, and mixed measurement units - conditions frequently encountered in agricultural trial reports. Broader reviews of PDF extraction workflows similarly emphasize the need for multi-stage normalization pipelines that enforce consistency across heterogeneous documents [8], [9].

Parallel research on retrieval-augmented generation (RAG) has demonstrated that combining dense retrieval with LLM reasoning significantly improves factual grounding and context relevance [1], [2]. Nevertheless, existing RAG systems primarily operate over unstructured text and lack mechanisms for numeric grounding or structured comparison. Evaluations of RAG frameworks show limitations when dealing with domain-specific metrics or analytical tasks requiring schema-aware reasoning [5]. Recent literature on hybrid retrieval for scientific navigation suggests integrating structured data sources into RAG pipelines to address these weaknesses [10].

AgriChat builds directly on these insights by combining multi-stage PDF normalization with hybrid structured–unstructured retrieval. Unlike prior systems that address document extraction or semantic retrieval in isolation, AgriChat provides an end-to-end solution capable of handling numeric comparisons, table normalization, metadata reconstruction, and semantic reasoning within a unified framework tailored to agricultural analytics.

III. Methodology

This section describes the end-to-end design and implementation of AgriChat, from raw PDF ingestion of agricultural trial reports to structured data normalization, embedding and indexing, and hybrid retrieval-generation mechanisms.

A. Overview of Pipeline Architecture

AgriChat implements a multi-stage pipeline: (1) PDF parsing to page-level JSON, (2) block-level flattening (bronze layer), (3) table normalization to structured rows (silver layer), (4) natural-language (NL) serialization of rows and unstructured text, (5) embedding generation and vector store indexing, (6) hybrid retrieval and LLM-based reasoning for user queries. This layered architecture ensures both **structured data fidelity** (for numeric metrics) and **semantic search flexibility** (for free-text queries).

B. PDF Parsing & Preprocessing

1) Page-level Extraction: A layout-aware text extraction library is used to parse raw PDF pages, extracting text blocks (paragraphs) and detecting tables. To guard against diverse PDF layouts and inconsistent table formatting, the system optionally supports table extraction via tools such as Camelot or Tabula when enabled, thus paralleling best practices from recent deep-learning and ML-based table extraction studies [6], [7].

For each page, the parsed output includes: `text_blocks`, `tables` (if any), PDF basename, page number, optional metadata (e.g., year, section type), and provenance information.

2) Flattening to Bronze Layer: Each extracted page JSON is flattened such that each row in the "bronze" dataset corresponds to either one text block or one table. For tables, both column header definitions and raw table rows (serialized as JSON) are preserved in the block record. This design ensures that all content is represented uniformly in a tabular, parquet-based staging layer, facilitating downstream normalization and reproducible indexing.

This two-stage extraction and flattening approach mirrors recognized paradigms in large-scale document ingestion systems [8], [9].

C. Table Normalization & Structured Data Construction

Recognizing that agricultural trial tables exhibit widely varying header names, units, and formats, AgriChat performs robust normalization:

- **Header alignment / promotion**: Header-like strings in early rows are promoted as true column labels.
- **Column detection via regex**: Key metrics such as yield, turnout, micronaire, length, strength, and uniformity are identified across naming variations.
- **Value cleaning and range validation**: Numeric fields are parsed to floats; physically implausible values are filtered out.
- **Entity normalization**: Variety names and categorical fields are cleaned to reduce noise (e.g., removal of statistical tokens such as "Mean", "CV").
- **Metadata inference**: Missing county/region fields are inferred from contextual text using regex.
- **Citation provenance**: Each normalized row stores PDF name, page, and original table identifier.

The resulting structured dataset is stored in parquet format with standardized fields: variety, county, year, yield, turnout, micronaire, length, strength, uniformity, loan value, lint value, and provenance metadata. This enables reliable downstream analytics and supports numeric comparative queries.

D. Embedding Generation and Vector Indexing

To support semantic search beyond structured numeric queries, AgriChat employs a dual embedding strategy:

- **Natural-language serialization**: Both text blocks and structured rows are converted to concise human-readable sentences (e.g., "County: Lubbock — Year: 2022 — Variety: NG 4190 B3XF — Yield: 1325 lbs/ac — Turnout: 38.5% ...").
- **Domain-adapted embedding model**: A dense encoder tuned for agricultural and numeric semantics produces embeddings for vector similarity search.
- **Persistent vector store**: Embeddings and metadata are indexed in a scalable retrieval framework to support low-latency document lookup.

This embedding-indexing approach draws on principles of retrieval-augmented generation (RAG), combining dense retrieval strengths with structured data grounding [1], [10].

E. Hybrid Query Routing and LLM-Backed Reasoning

A key innovation in AgriChat is the **dual-mode query router**, which interprets user intent and selects the appropriate data access strategy:

- **Structured-data mode**: Pattern-based routing ensures direct dataframe filtering for precise numeric comparisons when specific varieties, counties, or years are referenced.
- **Fallback RAG mode**: For open-ended queries, top-k embedding results are retrieved and provided to an LLM for context-aware generation.
- **Batch summarization**: Large result sets are summarized in multi-stage batches to ensure coverage without exceeding LLM context limits.

The pipeline integrates a high-throughput LLM backend optimized for both latency and factual consistency.

F. Implementation Details and Reproducibility

- The pipeline is modular, with separation of parsing, normalization, embedding, indexing, retrieval, and generation.
- Provenance metadata is preserved end-to-end for transparent traceability.
- The vector store supports incremental ingestion of newly published trial reports.
- Data quality safeguards ensure robustness against malformed or inconsistent input documents.

G. Relation to Prior Work

The methodology aligns with best practices in recent document-intelligence research. Automated extraction from scientific PDFs using hybrid OCR and deep learning approaches has become increasingly effective. Modern RAG systems integrate structured data pipelines to reduce hallucination and improve domain performance. Additionally, recent experience reports highlight similar workflows and challenges when constructing PDF-driven RAG systems, underscoring the relevance of AgriChat's design.

IV. System Functionality and Working of the Project

This section describes the operational workflow of AgriChat, detailing how user queries move through the hybrid structured–unstructured retrieval pipeline.

A. Unified Ingestion and Indexing Framework

AgriChat operates on a unified ingestion layer that integrates multi-stage PDF parsing with vector-based retrieval. The workflow consists of:

1) **Raw Extraction:** Each PDF page is parsed into text blocks, tables, and metadata using PDFMiner. Outputs are stored as JSON files with layout and provenance information.
2) **Bronze Layer:** All extracted blocks are flattened into uniform rectangular structures, stored in Parquet format to support reproducible preprocessing.
3) **Silver Layer:** Headers are inferred, inconsistent column names normalized, numeric values cleaned, and entities such as varieties and counties standardized.
4) **Natural-Language Serialization:** Each structured row is transformed into a concise descriptive sentence, enabling dense embedding models to capture semantic relations.
5) **Embedding and Indexing:** Using BGE-small-en-v1.5, both structured and unstructured sentences are embedded and stored in a LlamaIndex-managed persistent vector index with complete metadata bindings.

This unified representation allows AgriChat to retrieve both structured metrics and narrative context through a single vector store interface.

B. Intelligent Query Router

During inference, user queries are evaluated by an intent classifier combining regular expressions, domain keyword rules, and LLM-assisted fallback. Queries are routed to one of three modes:

1) **Structured Statistical Mode (Path B):** Activated when queries involve numeric metrics (e.g., yield, micronaire) or comparative expressions. Entities are extracted and translated into SQL queries executed on Supabase PostgreSQL. Results may be visualized using Plotly.
2) **Batch Summarization Mode:** Triggered by queries with summarization intent. Relevant blocks are retrieved in batches and summarized using Groq-accelerated Llama 3.3 while avoiding context-window overflow.
3) **Semantic RAG Mode (Path A):** Used for narrative, open-domain, or contextual queries. Top-k chunks are retrieved, context windows reconstructed, and responses generated by Llama 3.3.

C. User Interface Integration

A Streamlit interface exposes the full pipeline, allowing users to:

- toggle between RAG and structured modes,
- adjust the top-k retrieval parameter,
- preview SQL results and DataFrames,
- generate comparison tables and visualizations,
- receive coherent, grounded summaries from the LLM.

This interface unifies both retrieval paths, providing transparent and interactive access to agricultural analytics.

V. Conclusion

AgriChat introduces a hybrid structured–unstructured retrieval framework for agricultural variety trial analysis. By integrating multi-stage PDF normalization, metadata reconstruction, structured SQL querying, and semantic RAG retrieval, the system overcomes key limitations of both purely structured and purely unstructured methods. Experimental results demonstrate that AgriChat substantially improves retrieval consistency, reduces hallucination, and enables accurate cross-county and cross-year comparisons of agronomic performance metrics.

The system provides a scalable foundation for agricultural decision-support tools and illustrates the value of combining structured analytics with modern retrieval-augmented LLMs. Future extensions may incorporate additional crop types, expand the schema to support more fiber-quality attributes, or integrate fine-tuned domain-specific LLMs to further enhance reasoning accuracy.

GitHub Repository: https://github.com/kaushiks5499/AgriChat

References

[1] T. Gao *et al.*, "Retrieval-augmented generation for large language models: A survey," *arXiv preprint arXiv:2404.10981*, 2024. [Online]. Available: https://arxiv.org/abs/2404.10981

[2] W. Zhang *et al.*, "Rag: Enhancing retrieval-augmented generation for expert domain documents," *arXiv preprint arXiv:2312.10997*, 2023. [Online]. Available: https://arxiv.org/abs/2312.10997

[3] D. Prasad *et al.*, "Deep learning for table detection and structure recognition in document images: A review," *arXiv preprint arXiv:2206.05363*, 2022. [Online]. Available: https://arxiv.org/abs/2206.05363

[4] B. Smock *et al.*, "Pubtables-1m: Towards comprehensive table extraction from unstructured documents," *arXiv preprint arXiv:2110.04627*, 2022. [Online]. Available: https://arxiv.org/abs/2110.04627

[5] P. Lewis *et al.*, "Holistic evaluation of retrieval-augmented generation systems," *arXiv preprint arXiv:2309.05930*, 2023. [Online]. Available: https://arxiv.org/abs/2309.05930

[6] L. Sheng and S.-S. Xu, "Pdftable: A unified toolkit for deep learning-based table extraction," *arXiv preprint arXiv:2409.05125*, 2024. [Online]. Available: https://arxiv.org/abs/2409.05125

[7] G. S. Loitongbam and S. E. Middleton, "Tabular context-aware optical character recognition and tabular data reconstruction for historical records," *International Journal on Document Analysis and Recognition*, vol. 28, no. 3, pp. 357–376, 2025. [Online]. Available: https://link.springer.com/article/10.1007/s10032-025-00543-9

[8] J. Ramos *et al.*, "A review on knowledge and information extraction from pdf documents and storage approaches," *ResearchGate Preprint*, 2024. [Online]. Available: https://www.researchgate.net/publication/395310459

[9] K. Feldhoff, H. Wiemer, P. Träger, R. Kühne, M. Zimmermann, and S. Ihlenfeldt, "Automatic information extraction from scientific publications based on the use case of additive manufacturing," *Applied Sciences*, vol. 15, no. 17, p. 9331, 2025. [Online]. Available: https://www.mdpi.com/2076-3417/15/17/9331

[10] K. K. Ahmet Yasin Aytar, "A retrieval-augmented generation framework for academic literature navigation in data science," *arXiv preprint arXiv:2412.15404*, 2024. [Online]. Available: https://www.researchgate.net/publication/387321380$_{A}{}_{R}etrieval - Augmented_{G}eneration_{F}ramework_{f}or_{A}cademic_{L}iterature_{N}avigation_{i}n_{D}a$